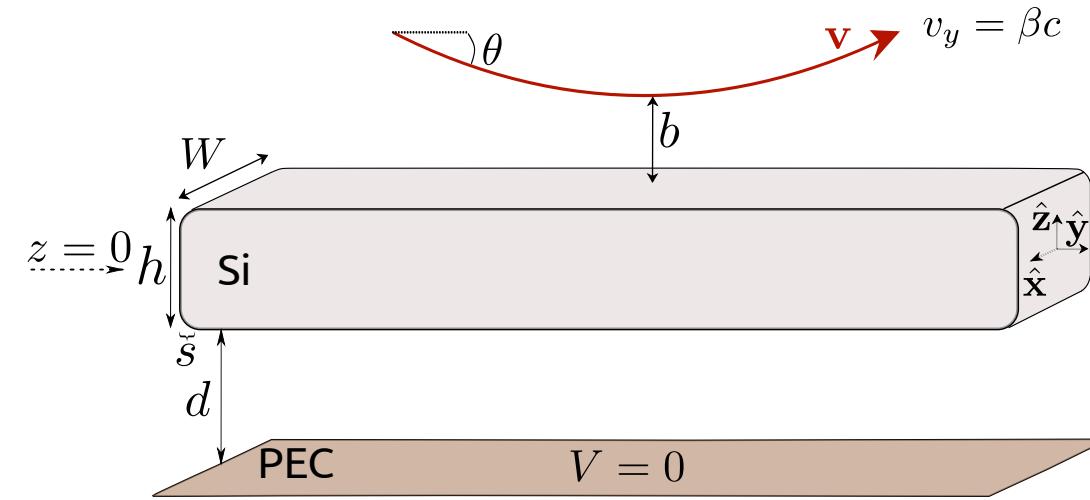


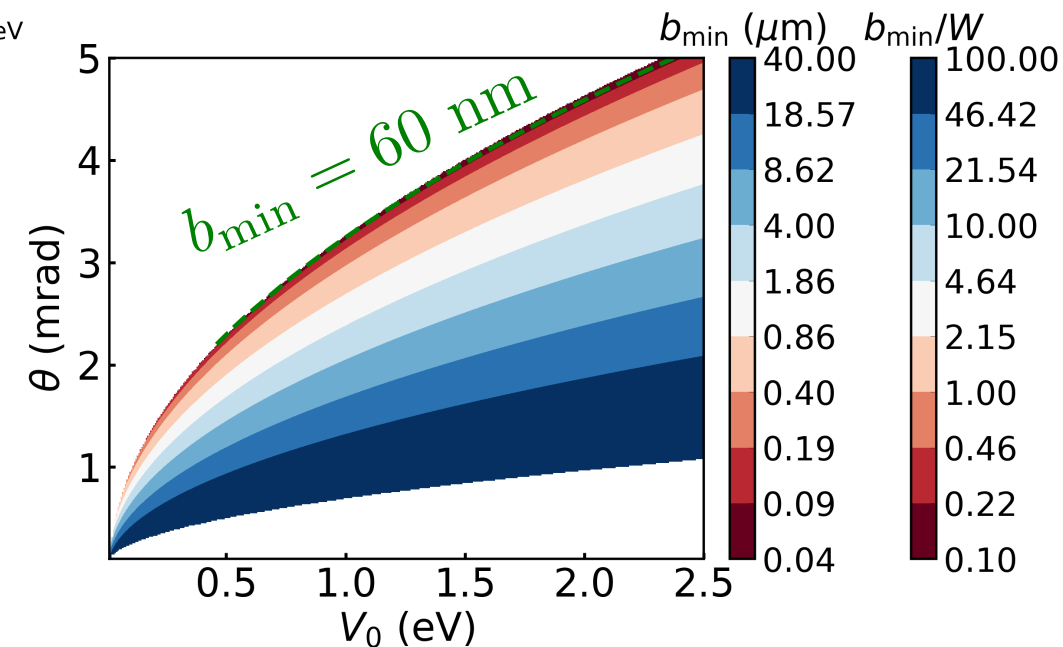
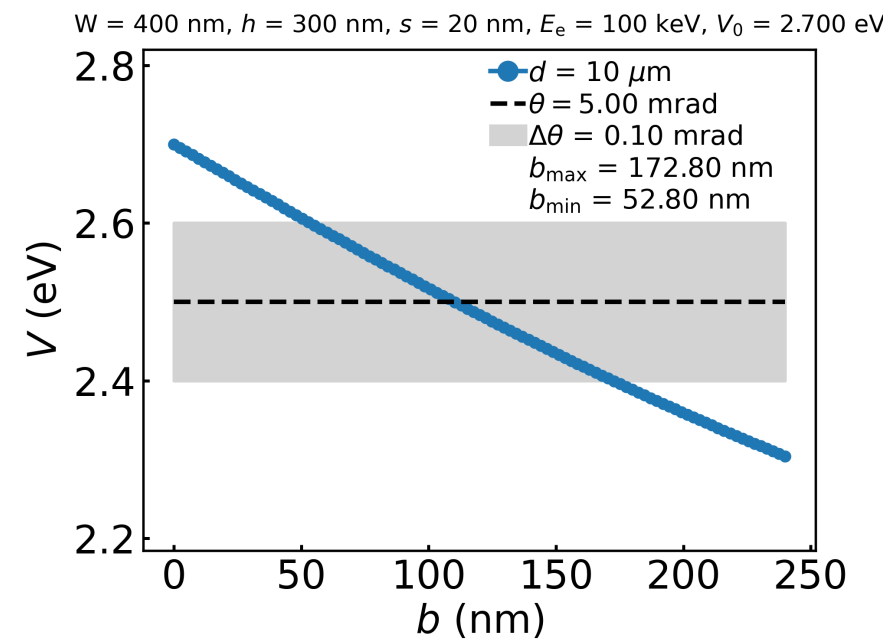
Electron-coupling-to-waveguide



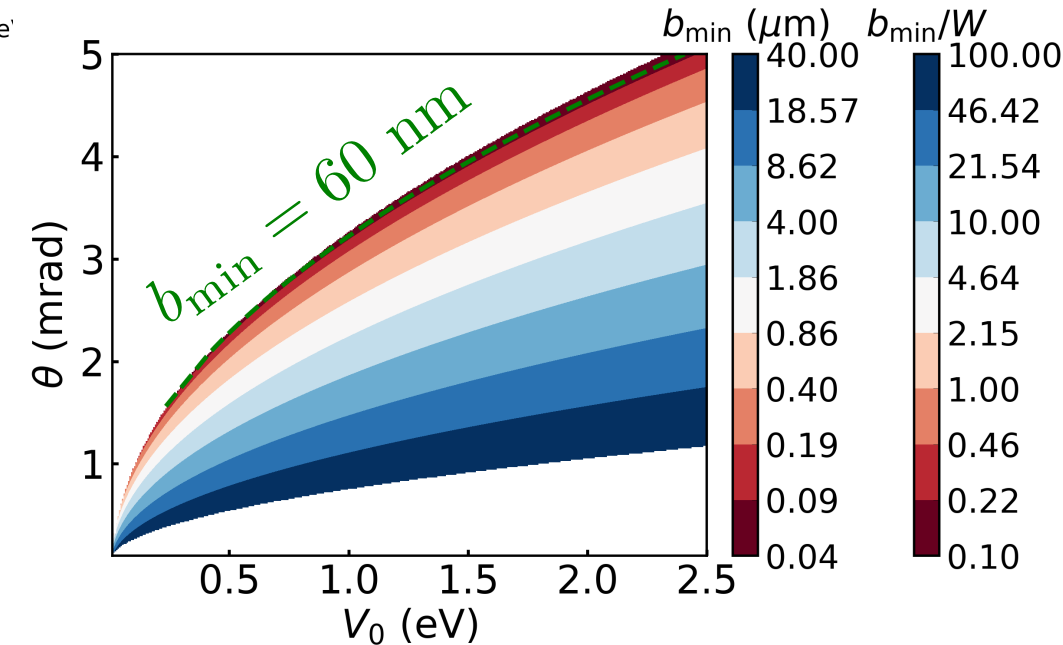
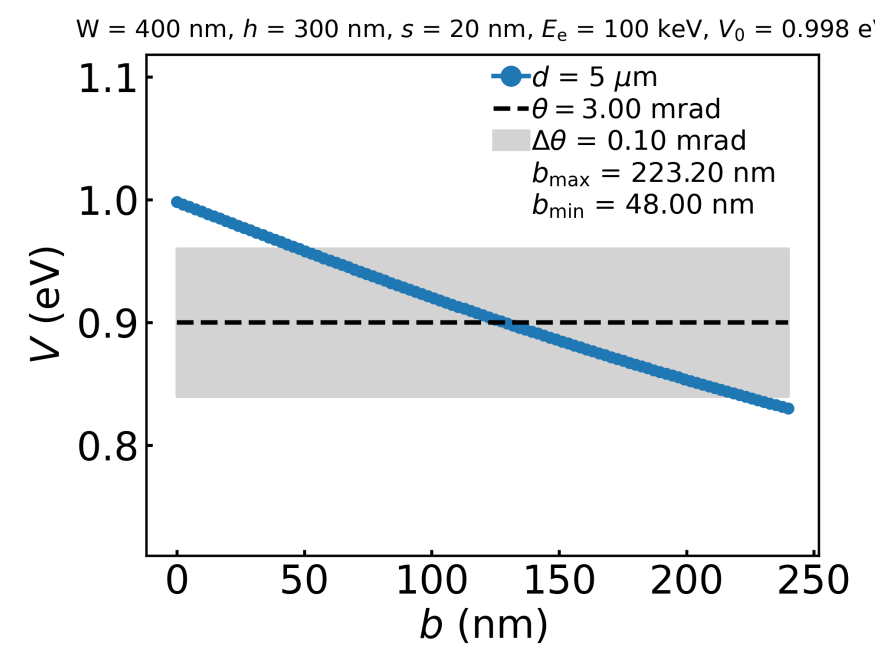
$$E_{\perp} = E_{\text{electron}} \theta^2 \text{ marked in dashed black line}$$

$$\Delta E_{\perp} = 2E_{\text{electron}} \theta \Delta\theta \text{ marked in grey}$$

$d = 10 \mu\text{m}$



$d = 5 \mu\text{m}$

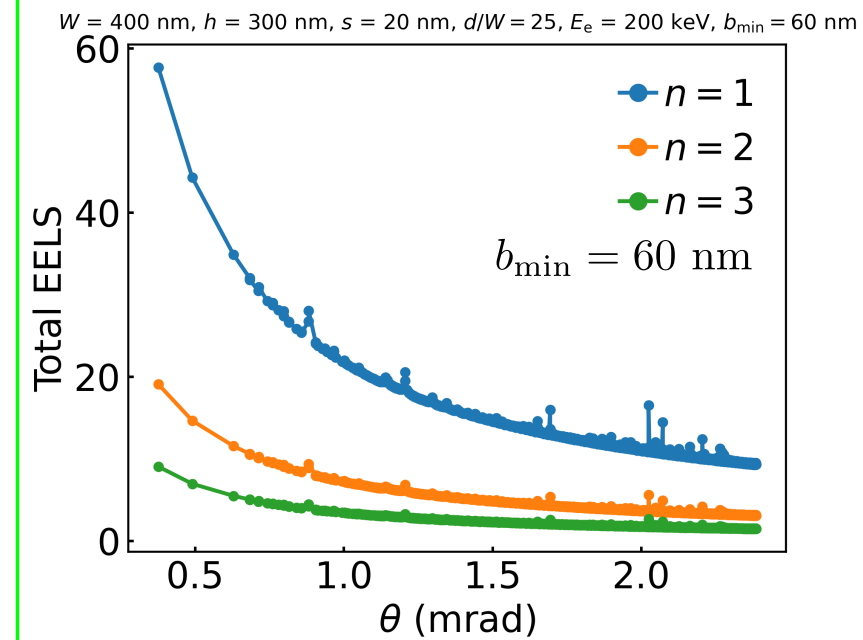


$$E_e = 100 \text{ keV} \quad W = 400 \text{ nm} \quad h = 300 \text{ nm}$$

Correction of the trajectory and integration

$$\Gamma_{\text{EELS}}(\omega) = 2 \int_{z_{\min}}^{\infty} \frac{vdz}{\sqrt{\left| c^2 \beta^2 \sin^2 \theta + \frac{2eV(z)}{m_e \gamma_e} \right|}} \frac{d\Gamma_{\text{EELS}}(\omega, z)}{dy}$$

EELS for different modes



EELS for different b for n = 1

